

Uncertainty in Machine Learning - WBAI054-05

Assignment 2

Total points: **40**
 Starting date: May 7, 2026
 Submission deadline: **23:59, May 21, 2026**

Lecturer Dr. Matias Valdenegro
General guidelines:

- The tasks are targeted at groups of 2-3 students. Please make sure that the load is well divided: every student should contribute. In the case of not equal contributions that grade will be divided accordingly.
- Please take advantage of the tutorial sessions to ask your questions about the tasks.
- Provide a (short but comprehensive) explanation of what you are doing for each task.
- A reviewer should be able to understand plots independently; be sure to label axes, a legend for colors, use an easily readable font size, etc.
- Refer to all plots, tables, code blocks, etc. in your report.
- Submit the report as a single PDF file through Brightspace.
- Submit all code in a zipped file named *Code_Group_GroupNumber*, e.g., *Code_Group_20* through Brightspace.

Uncertainty in Large Language Models

Maximum obtainable points: **20**

Large Language Models technically produce a probability distribution over tokens at each sequence element. Think about the following question:

Does this mean LLMs are performing uncertainty quantification?

Discuss whether token probabilities can be interpreted as calibrated confidence estimates, and what limitations exist in treating them this way. Consider the cases for hallucinations, incorrect predictions, and connections with the lecture on "Uncertainty Quantification methods".

Disentangling Classification Uncertainty

Maximum obtainable points: **20**

In the lectures we have seen the Gaussian logits formulation for classification uncertainty disentanglement. There is another formulation that is based on mutual information $\mathbb{I}(Y; \Theta)$:

$$\underbrace{\mathbb{I}(Y; \Theta)}_{\text{Epistemic}} \approx \underbrace{\mathbb{H}[\mathbb{E}_{\Theta}[p(y|x, \theta)]]}_{\text{Total}} - \underbrace{\mathbb{E}_{\Theta}[\mathbb{H}[p(y|x, \theta)]]}_{\text{Aleatoric}}. \quad (1)$$

Where $\mathbb{H}$ is the entropy, $\mathbb{H}[\mathbb{E}_{\Theta}[p(y|x, \theta)]]$ is the entropy of the mean probability and it is considered to represent the total uncertainty, the Expected Entropy of each probability vector $\mathbb{E}_{\Theta}[\mathbb{H}[p(y|x, \theta)]]$

represents aleatoric uncertainty, and the difference between them is considered a measure of epistemic uncertainty and is an approximation of the Mutual Information $\mathbb{I}(Y; \Theta)$, where Y is the Random Variable from which the label y is drawn.

Note that the symbol $\approx$ means approximate. More information can be found in the paper below.

Mucsanyi, Balint, Michael Kirchhof, and Seong Joon Oh. "Benchmarking uncertainty disentanglement: Specialized uncertainties for specialized tasks." *Advances in neural information processing systems* 37 (2024):

Which can also be found [here](#).

Using the information given above, make a comparison between the two uncertainty disentanglement formulations, considering the following factors: assumptions, and quality of uncertainty disentanglement, meaning making arguments on why these formulations might or might not work to disentangle aleatoric from epistemic uncertainty.

The arguments should be simple, based on the assumptions that you described.